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ORB-SLAM2 System Enhanced with SiaT-Hough Module on Mars Digital Twin Platform: For Rock Counting and 3D Reconstruction in Martian Environments

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ABSTRACT This paper proposes a digital twin platform for Mars environment based on NVIDIA Isaac Sim to create a unique random environment with multi-scale fine-grained terrain features through programmed terrain generation and asset management. The SiaT-Hough module was added to the ORB-SLAM2 system for Mars rock segmentation and 3D modeling, as well as rock counting. We propose a lightweight Siamese Transformer (SiaT) network combined with Hough transform for small sample semantic segmentation tasks, especially for rock counting in Mars exploration scenarios. Experimental results demonstrate that the SiaT-Hough model achieves an average segmentation accuracy of 87.86% and a processing speed of 57.34 FPS on the Mars simulation dataset, striking an optimal balance between accuracy and real-time performance. In addition, we also carried out 3D reconstruction positioning accuracy evaluation and point cloud quality evaluation, and the positioning absolute track error on the two public data sets was less than 0.1m, 0.0639m and 0.0106m, respectively. The Pearson linear correlation coefficient of mean point cloud mass in Mars rendering dataset reaches 0.852, and the mean relative error of rock count is 8.63%, which proves the feasibility of the improved ORB-SLAM2 system for rock count in Mars environment.

INDEX TERMS Mars Exploration, digital twin, ORB-SLAM2, Semantic segmentation

I. INTRODUCTION

Currently, NASA's Perseverance rover successfully landed on Mars on February 18, 2021, on a mission to search for signs of life. [1] The Zhurong rover on board China's "Astronomical 1" probe also made a successful soft landing on May 15, 2021, and has begun its exploration. [2] These missions mark a new wave of Mars exploration. However, the success rate of Mars exploration missions remains below 50%, mainly due to the complexity and extreme conditions of the Martian environment, including a thin and unstable atmosphere, low pressure, rugged terrain, frequent dust storms, significant temperature changes, and weak sunlight. In addition, the frequency of human exploration is limited, resulting in a relative scarcity of environmental sample data on Mars. These environmental challenges pose significant obstacles to the design, optimization, and upgrading of Mars exploration missions.

Digital twins integrate physical models, sensor data, and operational history to enable dynamic simulations of Mars rovers, characterized by three key capabilities: life-cycle tracking, real-time/near-real-time updates, and bidirectional mapping between virtual and physical systems. At present, a variety of Mars simulators have been developed for navigation, SLAM, mission planning and other functional tests. Earlier tools such as PANGU supported image generation of the planet's surface [3], and McCrum built on this by adding Martian rocks and dynamic atmospheric effects; Sun and Jiang developed a high-resolution terrain simulator and a modular terrain construction framework respectively [4,5].MarsSim based ROS/Gazebo platform [6], limited by rendering capabilities, is difficult to meet the needs of image recognition. The current research progress is slow, mainly limited by the rendering capability, compatibility and flexibility of the simulator. To solve these problems, more and more

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scholars have turned to the Isaac Sim platform based on Nvidia Omniverse [7,8]. Isaac Sim not only provides built-in sensor support and compatibility with common rover formats (such as URDF, SDF), but also has powerful ray tracing rendering capabilities that can handle large-scale scenarios in real time, significantly improving the fidelity and practicality of the simulation, and providing an ideal platform for efficient simulation verification of Mars exploration missions.

In the execution of vision-based 3D reconstruction of the Martian environment and landmark recognition tasks related to SLAM algorithm in low-feature outer space environments such as Mars, synthetic data generation has become a common means due to limited real data [9,10]. Currently, the publicly available Mars data sets have different focuses. GMSRI helps to study the Mars environment based on texture [11], S5Mars provides semantic segmentation terrain data to assist the navigation of Mars exploration rovers [12], and AI4MARS is used to train and verify terrain classification models [13], but they are all fixed data sets and cannot generate new data. In addition to the Mars dataset, there are also publicly available lunar image datasets, such as the Artificial Lunar Landscape Dataset, ReSyRIS [14] and the blender based OAISYS[15], which focus on terrain classification. ReSyRIS and blender based OAISYS focus on semantic segmentation, and both lack pixel-by-pixel instance masks, limiting their application in instance awareness tasks. In order to simulate the data set closer to the original Mars environment, we can integrate multi-source data, and combine various types of data obtained by the Mars probe with similar environmental data on the moon. Utilizing advanced terrain generation algorithms, such as generating Martian terrain based on high-resolution terrain data and physical models; Random factors in the Martian environment, such as random distribution of rocks and changing lighting conditions, are simulated by randomization method, so as to build a more realistic Mars environmental dataset.

As a key technology in Mars exploration, Mars rock segmentation is crucial for the autonomous navigation, path planning, and geological research of rovers. Accurate rock segmentation in Mars images helps rovers avoid obstacles and ensures safe travel. Early methods relied on traditional image - processing techniques. For instance, multi - scale feature fusion methods segment images into locally consistent units via superpixel segmentation and then merge regions based on multi - scale features[16]. However, methods like those based on sparse background modeling and regional contrast enhancement[17] have limited ability to detect small rocks on the Martian surface. With the rapid development of computing technology and machine learning, deep learning has been widely applied in Mars rock segmentation. Models such as SegMarsViT[18]and Light4Mars[19]reduce computational complexity through lightweight design. Yet, their segmentation accuracy for small - scale rock targets or sparsely annotated data is restricted by insufficient contextual information and pixel - level sparse annotations. RockFormer, a U - shaped Transformer framework, combines shallow detail and deep semantic information with its hierarchical encoder - decoder structure for multi - scale feature adaptation to Mars rocks' diverse shapes [20,21]. The high computational complexity of Transformer modules leads to a large model size, posing challenges for lightweight deployment in Mars exploration tasks due to computational resource constraints.

Current Mars simulators and rock segmentation techniques face several key challenges. These include the limited rendering capabilities, compatibility, and flexibility of simulators. Existing datasets are often restricted and lack pixel - level annotations. Additionally, rock segmentation algorithms struggle with accuracy, especially for small - scale targets and in complex environments, due to insufficient contextual information and sparse pixel - level annotations. There's also a significant contradiction between the large number of model parameters and the limited computational resources available for Mars exploration tasks. The limitations of current simulators and segmentation algorithms underscore an urgent demand for innovative solutions: computationally efficient models for real-time processing on edge devices, and high-fidelity simulation platforms to advance Mars exploration missions. To overcome the above problems, the work proposes a new open source robotic simulator for the Mars environment and adds the SiaT-Hough module to the ORB-SLAM2 system for Mars rock segmentation, 3D modeling, and point cloud generation. We provide a basic algorithmic framework for environment generation, as well as automatically generated annotated synthetic data sets.

Our contributions in this article are:

1. This paper proposes a Mars simulation method based on the Isaac platform to create a unique random environment with multi-scale fine-grained terrain features through programmed terrain generation and asset management.

2. This paper also proposes a new small sample instance segmentation network Siat-Hough model. We use the Hough transform guidance attention mechanism to generate edge weight graphs by using circular or straight line detection, and integrate them into the cross attention layer of SiaT, forcing the model to focus on regions with significant geometric features such as rocks and cracks to enhance the consistency of edge segmentation. It provides efficient and accurate segmentation of unknown class targets for Mars exploration missions, adapting to the complex Martian environment and real-time processing requirements.

3. We used the improved ORB-SLAM2 system for 3D reconstruction of the Mars simulation environment,

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conducted positioning accuracy evaluation and point cloud quality evaluation, and finally performed rock counts.

II. MATERIALS AND METHODS

A. DATA ACQUISITION AND MODELING METHODS

A key component of the Mars digital twin involves physical modeling of the Martian environment. In order to generate highly realistic Martian geomorphological features, this section emphasizes the construction and analysis of Martian surface textures, rock distribution, program asset management, rover models, and the ORB-SLAM2 system. Through the above techniques and methods, we have created a high-fidelity Mars environment simulation framework. The framework achieves visual consistency with real Martian environments and generates high-definition semantic images via ORB-SLAM2 integration. A flowchart illustrating this process is shown in Figure 1.

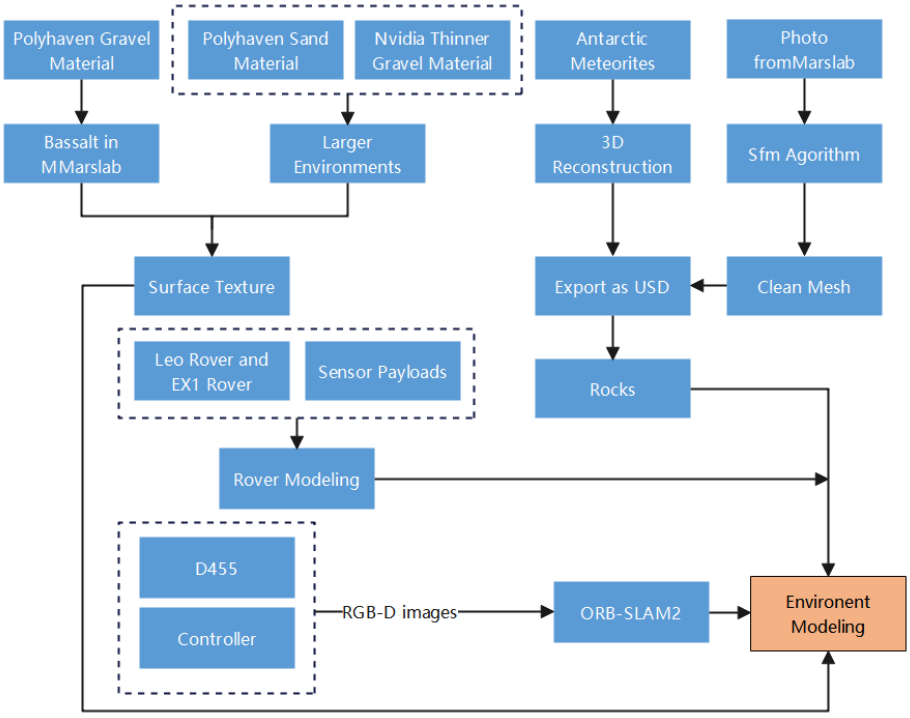

FIGURE 1. Mars Environment Modeling Process

B. EXPERIMENTAL PLATFORM

1) DIGITAL TWIN PLATFORM

In order to simulate the rover's kinematics and sensor payloads for vision tasks, it is necessary to create realistic models of the vehicle and sensors. Our simulator provides a rover model and a pre-configured payload sensor: the EX1 rover. To simulate payload sensors such as stereo cameras, appropriate camera parameters and attachment positions must be provided. As the target camera model, we chose to use Intel RealSense D455 because it is so widely used in the field of rovers. The parameters of the simulated camera are determined according to the actual sensor specifications. The rover's camera is mounted on the front end of the right rocker arm link 0.25 meters above the ground, as shown in Figure 2. The following is a general schematic of the rover's appearance, its overall design is compact and rugged, divided into sensing and data acquisition and calculation and positioning equipment.

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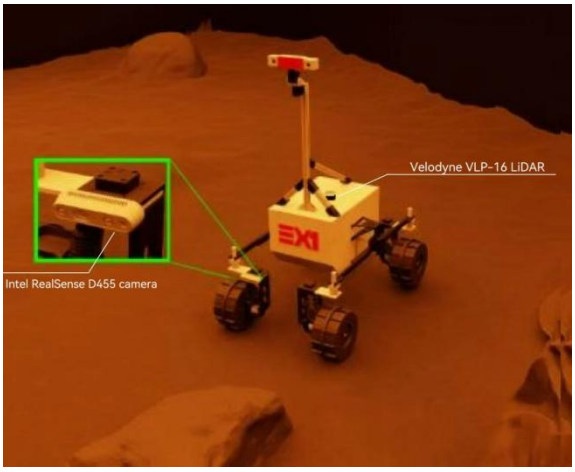

FIGURE 2.EX1 rover: Equipped with sensors in the Marslab environment

In terms of sensing and data acquisition, Intel RealSense D455 camera is used to obtain RGB-D image data in real time, which can be used for terrain texture analysis, small obstacle detection and visual SLAM construction. At the same time, the Velodyne VLP-16 liDAR is used to generate high-density three-dimensional point clouds, enabling Mars rock edge identification, large-scale terrain modeling, and path planning obstacle detection. In terms of computation and positioning, NVIDIA Jetson AGX Xavier uses real-time rock segmentation networks, multi-sensor data fusion, and autonomous navigation decisions. With the help of RTK positioning module, centimeter-level absolute positioning is realized based on the simulated Mars reference station network, and multi-source fusion positioning framework is formed with SLAM system.

2) SOFTWARE PLATFORM

The Marslab platform is a synthetic data generation and simulation environment built based on NVIDIA Isaac Sim for Mars rock research. With NVIDIA's advantages in graphics processing and simulation technology, it provides a powerful help for Mars environment simulation. It has the function of programmed terrain generation, which can generate randomly distributed rock terrain based on the Poisson point process, simulate different scenarios by adjusting parameters, and ensure data repeatability by combining randomization. Integrate efficient synthetic data annotation tools to automatically add accurate annotation to image and point cloud data, covering boundary boxes, instance masks and various morphological parameters, compared with manual annotation, faster, more efficient and more accurate; It can also conduct multi-physics simulation, using the NVIDIA Omniverse path tracking engine to simulate physical phenomena such as Martian illumination and surface reflection, and consider the impact of temperature and atmospheric scattering on sensor data to generate highly realistic synthetic data. In addition, the platform has technical advantages such as high-fidelity simulation, good scalability, and seamless integration with deep learning frameworks.

At work, Marslab platform first sets points' density, distribution range and other parameters in the programmed terrain generation module based on the Poisson point process according to research requirements, and combines seed randomization to determine the initial conditions to generate Mars terrain with different rock distribution characteristics. Then, the multi-physics simulation module is used to simulate physical phenomena such as light, temperature and atmospheric scattering on the surface of Mars through the NVIDIA Omniverse path tracking engine, and these physical characteristics are integrated into the generated terrain scenes to build a highly realistic Mars environment. In the data generation stage, images and point cloud data in this environment are collected by various virtual sensors. At the same time, the synthetic data annotation tool is started to automatically add annotation information such as boundary boxes, instance masks and rock morphology parameters to the acquired images and point cloud data according to the pre-set annotation rules. Finally, the generated synthetic data with precise annotations is output in a uniform data format through a seamless interface with deep learning frameworks such as PyTorch for researchers to use directly for model training and optimization.

C. SIMULATION ENVIRONMENT GENERATION METHOD

The following discussion covers how to generate and manage terrain in our simulator. When building a Mars simulation environment, high-quality Mars terrain simulation is an indispensable core element. The process of terrain generation and rendering begins with data collection from a laboratory environment. A total station scans the ground to produce point clouds, which are then processed into a Digital Elevation Model (DEM). At the same time, real Mars terrain data is acquired, and the Ames Stereo Pipeline (ASP) and images from the Tianwen-1 mission are used to create a large-scale DEM. Next, feature parameters are extracted from the Mars mission DEM data for analysis. Unique random environments with fine - grained features are generated through a procedural approach, and a seeded randomizer is used to ensure simulation consistency. The generated terrain is then imported into IsaacSim, where the DEM is dynamically imported and randomized by editing the mesh properties of the USD stage. Finally, IsaacSim's rendering engine is used for high-quality ray tracing or path tracing rendering, and procedural asset management is applied to optimize rendering performance. This completes the terrain generation and rendering process, constructing a high - fidelity Mars simulation environment. The workflow diagram is shown in Figure 3.

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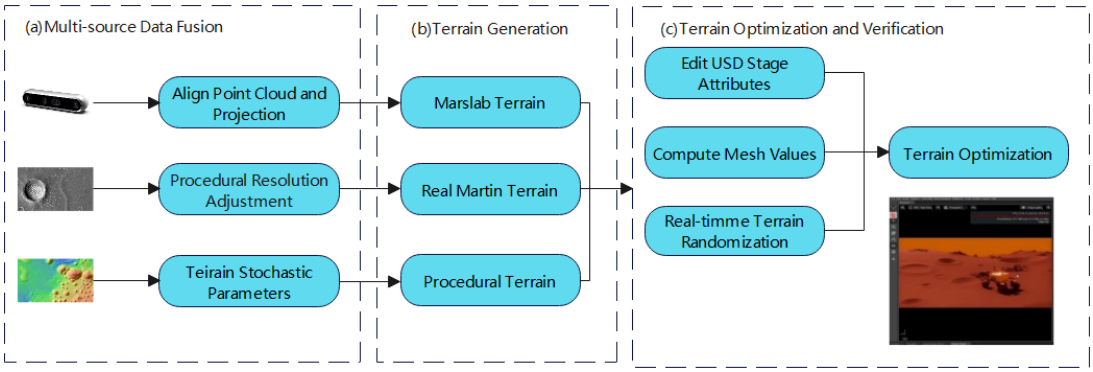

FIGURE 3. The process of terrain generation and rendering

1) TERRAIN GENERATION AND RENDERING

Data Source

HIRISE high-resolution Mars DEM (resolution 1 m) and the three-dimensional image of TiAN-1 mission are used as the basic data for terrain generation to ensure that the generated terrain has real Martian geomorphic features. One environment in our simulator is a 4.5 m × 9 m Mars simulation facility at the University of Space Engineering (Figure 4). To create a digital twin of this laboratory, the total station was used to scan the ground part from three different angles, generate three point clouds, align, merge, and project onto a grid with a resolution of 1 centimeter per pixel, manually clean the artifacts, and fill the gaps with cubic function interpolation. The digital elevation model can be loaded and used in the simulator. For the real Martian terrain, a digital elevation model of 3 km × 3 km with a resolution of 1 meter per pixel is created by relying on ASP and Astro 1 mission images, and the resolution can be artificially reduced to 4 centimeters per pixel through procedural generation if necessary.

(a) MarsLab

(b) Detectors

FIGURE 4. Mars Environmental Laboratory scene

Programmatic Generation

The Poisson point process, a stochastic model generating spatially random yet uniformly distributed points, simulates rock distributions on the Martian surface by capturing natural randomness while maintaining statistical uniformity. The generation of each point is independent, and the distribution of points follows a Poisson distribution. The Poisson point process allows control the distribution characteristics of points by setting parameters. Because of its randomness, tunable parameters, computational efficiency, and compatibility with other algorithms, it is the preferred method for simulating the distribution of rocks in natural terrain this paper. In comparison, a uniform or normal distribution may be too regular or concentrated to capture the complexity of natural terrain well. Its probability density function is given by Equation (1).In the context, λ represents the point density , V is the volume of the distribution area, and n is the number of points within area V .

$$P(n) = \frac{(\lambda V)^n e^{-\lambda V}}{n!} \quad (1)$$

To generate random rock distributions based on the Poisson point process, we set different parameters. The point density can be 3 - 5 points per square meter. On average, 122 - 203 rocks randomly distribute across a 40.5 - square - meter area.To ensure that the rock distribution generated each time has diversity and experimental repeatability. We analyzed the DEM data of Astronomical 1, HiRISE and CTX missions, extracted the basic features of terrain generation, combined multiple terrain parameters, including plains, dunes, canyons and meteorites, in a specific order to generate multi-terrain structures, and combined the randomizer to ensure the consistency of simulation examples. Create unique random environments with fine-grained terrain features.

Rendering Technique

The NVIDIA Omniverse path-tracking engine accurately simulates Martian light and surface reflection characteristics, making the resulting simulated images more realistic. When importing a terrain in IsaacSim, the Mesh of the mesh is dynamically edited in the USD phase: FaceVertexIndices (connectivity of vertices), FaceVertexCounts (number of vertices), Points (position of vertices) and ST (UV coordinates of each element in FaceVertexIndices) because the digital elevation model represents a regular grid, The calculation of the values is intuitive, enabling randomization of the terrain

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without exiting the simulation, and flexibility for vehicles to leave traces on the terrain in the future.

2) IMPROVED ORB-SLAM2 SYSTEM

Considering the complexity of the Mars environment, we selected ORB-SLAM2, which performs well in the outdoor environment, as the baseline model for the three-dimensional reconstruction of the system, and added a semantic segmentation module to it. The flow chart of the improved model is shown in Figure 5, and the improved module is represented in orange. The ORB-SLAM2 system is divided into four parts: front end, back end, loop detection and drawing. First, the system initializes the pose of the camera by obtaining RGB-D image information, extracts feature points, and generates key frames according to the number of obtained feature points. Here, only the key frame image is semantically divided and dense semantic point cloud map is generated. At the same time, the keyframe is used as the basis of loopback detection thread to correct the accumulated error of the current dense point cloud map, and the final dense point cloud map is stored by Octomap.

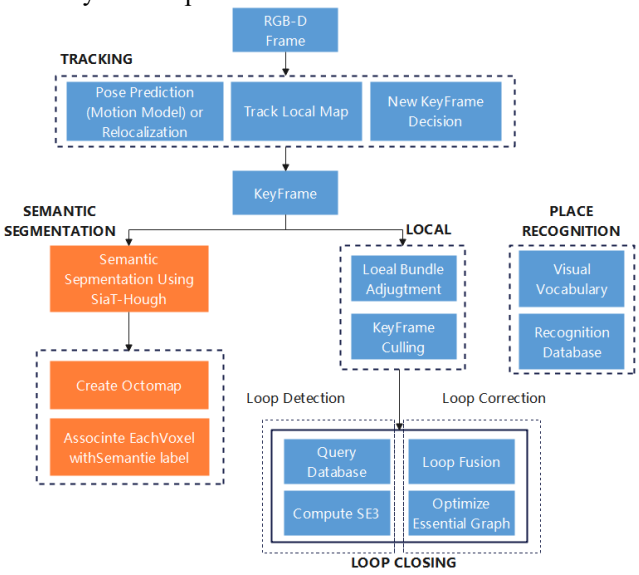

FIGURE 5.Flow chart of three-dimensional semantic reconstruction system

D. INSTANCE SEGMENTATION NETWORK

1) LIGHTWEIGHT SIAMESE TRANSFORMER NETWORK ARCHITECTURE

Lightweight Siamese Transformer (SiaT) is an efficient model designed for small sample semantic segmentation tasks, the core goal is to achieve accurate segmentation of unknown class objects with a very small number of labeled samples (such as 1-5 supporting images). In Mars exploration scenarios, SiaT's lightweight nature and powerful feature fusion capability make it ideal for processing complex terrain imaging data.As shown in Figure 6, the backbone feature extractor takes support images and query images as inputs to generate high - level and mid - level features. It uses high - level support and query features along with support masks to create an initial binary mask for query images. Then, a Siamese Transformer Module (STM) generates enhanced support and query prototypes, which are fed into a Query Activation Module (QAM) to interact with query features. The QAM - enhanced query features are integrated with edge weights, enabling the Hough Transform to detect geometric shapes more accurately. Finally, these features are input into an encoder to generate the final query segmentation masks.

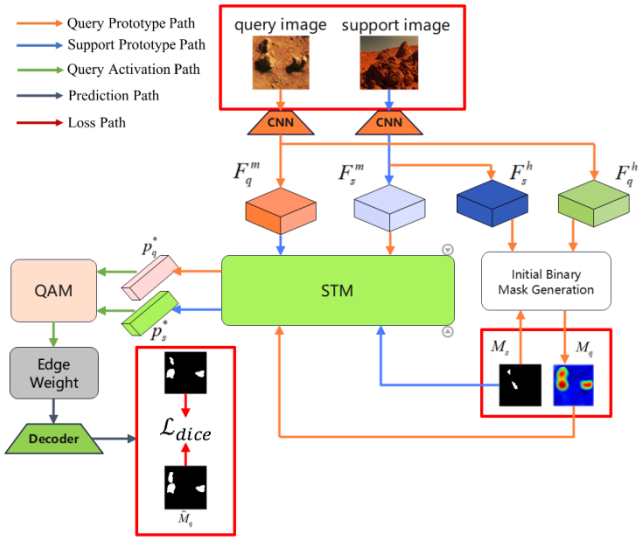

FIGURE 6.Overall architecture of SiaT. F_s^h, F_q^h and F_s^m, F_q^m indicate the high - level and mid - level features of support and query images, which are produced by the backbone network. M_s and M_q are the support mask and initial binary query mask, respectively. P_s^* and P_q^* are the enhanced support and query prototypes generated by the Siamese Transformer Module (STM). $\bar{M}_q$ represents the final segmentation result of the query image. The loss function measures the difference between the model's predictions and the true annotations by comparing the query image's segmentation result with the predicted segmentation of the support image.

1.Feature and initial binary mask

In our proposed framework, we utilize ResNets[22] with dilated convolutions pretrained on ImageNet[23]as the backbone feature extractor. Inspired by the approach in PFENet[24], we extract high - level and mid - level features from the query and support sets. These features are used for subsequent feature enhancement related to masks M_s and M_q. The high - level features F_s^h and F_q^h, along with the support mask Ms, are used to create the initial binary query mask M_q. Its generation process follows the method in Ref. [24]. Specifically, we generate a prior mask M_1 that reflects the confidence of each pixel belonging to the target class. Higher values in M_1 indicate a higher probability of the pixel being in the target region. Then, we introduce a threshold τ to determine the query segmentation mask M_q as follows:

$$M_q = I(M_1 > \tau) \quad (2)$$

where I is the index function.

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2. Siamese Transformer Module (STM) : Enhanced support for image prototypes and query image prototypes through two Transformer decoders with shared parameters.

(1) Supporting prototype path: mask features (such as crater annotation) supporting images are taken as Query, combined with image features (Key/Value) to extract common target information.

(2) Query prototype path: Reverse use of query image features to generate prototypes, fusion of supporting image foreground information, and improve the adaptability to the dynamic changes of the Martian surface.

3. Query Activation Module (QAM) : dynamically assign enhanced prototypes to query features and generate high-resolution segmentation mask through channel splicing, which is suitable for fine identification of rocks, cracks and other targets.

The adaptability of SiaT network and Mars scene is highlighted in that SiaT can suppress the influence of dust noise and illumination changes in Mars images through global attention mechanism, and support freezing pre-training backbone network with few parameters, which is suitable for the real-time processing requirements of the Mars exploration rover edge computing platform (NVIDIA Jetson AGX Xavier). In Mars missions where labeling data is scarce, only a small number of labeled samples can be quickly adapted to rock groups.

2) HOUGH TRANSFORM

Hough transform is based on the principle of converting curves or shapes in the image space to parameter space for detection. In the image obtained by the Mars probe, Hough transform maps points that may form specific geometric shapes (such as the circle of the crater, the straight line at the edge of the canyon, etc.) to the parameter space through the analysis of pixels in the image. In the parameter space, the parameters of these shapes are determined through statistics and screening, and then the corresponding geomorphic shapes are accurately detected. These detection results provide landmark reference for the detector's positioning and navigation, and help the detector to determine its own position and plan a safe path to avoid obstacles. At the same time, the detected shape information can be fused with other probe data to optimize the construction of Mars surface maps and enhance the accuracy and completeness of the maps.

3) COMBINATION OF SIAT NETWORK AND HOUGH TRANSFORM

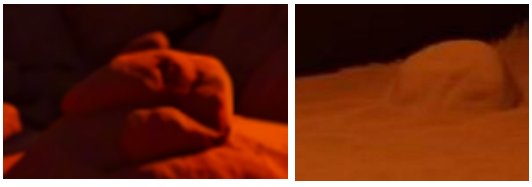
The combination of lightweight Siamese Transformer (SiaT) and Hough transform improves segmentation accuracy and efficiency in Mars exploration scenarios through a geometric-semantic dual-drive framework. First, the Hough transform directed attention mechanism uses circular or straight line detection to generate Edge Weight maps, which are integrated into the cross attention layer of SiaT, forcing the model to focus on regions with significant geometric features such as craters and cracks, and enhancing the consistency of edge segmentation. Secondly, cascaded region screening is used to pre-extract the candidate target region (such as the crater boundary frame) by Hough transform, and SiaT is only run on the local region to reduce the computational load. At the same time, combined with post-processing optimization, Hough transform was used to correct the fracture contour, complete the mask edge geometry, and improve the reliability and visual quality of the results.

III. EXPERIMENTS AND DISCUSSION

For such simulators, one of the most critical requirements is to minimize the gap between the simulation and the real world. This is crucial for perceptual tasks because models trained on simulated data may not generalize to the real world if they are not accurately modeled. To validate the simulator's ability to bridge the simulation-to-reality gap, we trained the neural network on both synthetic and real-world datasets under diverse environmental conditions. To achieve this, we utilize 3D mapping tasks to observe how various simulation Settings affect the performance of the network when applied to real-world data.

A. DATA ACQUISITION

To generate the real environment dataset, we used the RealSenseD455 camera to capture images of the inside of the actual Marslab.We only save RGB data with a frequency of 10Hz. We collected three rosbags in which we reserved one frame for every 15 consecutive frames, resulting in approximately 1207 images per rosbag. We manually labeled the rocks in each image. We used data from high and low light positions for the training set (825 images) and data from low light positions for the test set (382 images).



(a) Front (b) Back

FIGURE 7.Simulated Rock Images in MarsLab.The Mars rocks are constructed in a virtual environment, and the subsequent experiments are also carried out in this environment.

B. INSTANCE SEGMENTATION NETWORK COMPARATIVE EXPERIMENTS

In order to evaluate the performance of the proposed example segmentation network for rock segmentation, a geometric-semantic dual drive object detection model based on SiaT combined with Hough transform was used to detect various Mars rocks in different data sets, and compared with Mask-RCNN network. In order to verify the effectiveness of the proposed improvement strategy, the average accuracy (AP), equal-crossover ratio (MloU) and frames per second (FPS) were used for quantitative evaluation. Where the average accuracy can be expressed by formula (5), the parity

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ratio can be expressed by formula (6), where TP represents the number of true positives, FN represents the number of false negatives, FP represents the number of false positives, k represents the number of all classes except the background, and n represents the NTH class. The computer configuration in the experiment consisted of two NVIDIA TITAN Gpus, 24gb memory, Ubuntu18.04 operating system, Cuda10.0, Cudnn7.5 and Opencv3.5.6, and was trained using Keras framework.

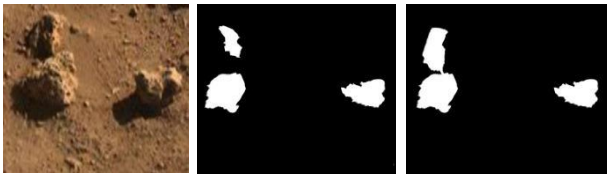
$$P = \frac{TP}{TP + FP} \tag{3}$$

$$R = \frac{TP}{TP + FN} \tag{4}$$

$$AP = \sum_{n=1}^{k} (R_n - R_{n-1}) \cdot P_n \tag{5}$$

$$MIoU = \frac{1}{k+1} \sum_{i=0}^{k} \frac{TP}{FN + FP + TP} \tag{6}$$

In this paper, the network model based on SiaT-Hough framework fusion segmentation module, to achieve the Mars rock case segmentation. Therefore, in the experimental stage, we need to combine the semantic segmentation model Mask RCNN, YOLOv5 model, pspnet, YOLOv8 model and the classic semantic segmentation model Deeplabv3 for comparative analysis. In order to highlight the true segmentation results of the image (a), we also show the predicted segmentation results of the support image (b) in Figure 8, which shows the segmentation results of the SiaT-Hough network, and Table 1 shows the evaluation results of different algorithms under the verification set. Figure 9 shows the segmentation results corresponding to six different networks.



(a) (b) (c)

FIGURE 8.Example of SiaT-Hough network segmentation result. (a) the input of the original RGB image,(b) represents the predicted segmentation results supporting the image and (c) represents the real segmentation results of the image. The segmentation results are divided into 2 categories, where white represents Martian rocks and black represents the background.

TABLE1
EVALUATION RESULTS OF DIFFERENT NETWORK MODELS

Model	Backbone	AP(PA) /%	mIoU/ %	FPS/(Frame•s^{-1})
YOLOv5	CSP-Darknet53	80.41	74.92	70.51
pspnet	ResNet50	80.50	74.81	42.63
DeepLabV3+	MobileNetV2	82.82	75.08	30.63
YOLOv8	CSP-Darknet53	83.94	76.01	75.36
Mask R-CNN	ResNet50	84.74	75.41	6.78
SiaT-Hough	**ResNet101**	**87.86**	**78.37**	**57.34**

TABLE2
EVALUATION RESULTS OF DIFFERENT MODULES

model	modules	backbone	AP (PA) /%	mIoU /%
SiaT-STM	STM	ResNet101	83.20	75.32
SiaT-STM-QAM	STM.QAM	ResNet101	85.60	76.42
SiaT-Hough	STM.QAM.Hough	ResNet101	87.86	78.37

As shown in Table 1, the SiaT-Hough model achieves an average accuracy of 87.86%, outperforming Mask R-CNN by 3.12%. Notably, its lightweight architecture requires only 0.68M parameters, significantly reducing computational complexity compared to ResNet-based backbones (e.g., 25.6M in Mask R-CNN). This significantly boosts the segmentation speed, making the FPS 8.5 times higher than that of Mask R-CNN. Compared to YOLOv5 and PSPNet, the segmentation accuracy is substantially improved. However, the ResNet101 model's more complex structure than CSP-net53 leads to a slight decrease in FPS. Despite differences in the six segmentation networks' architectures, their mIoU values are relatively close. SiaT-Hough has the highest mIoU of 78.37%. AP/PA values are crucial for evaluating target recognition and prediction accuracy and are equivalent in semantic segmentation. Compared to DeepLabV3+ with a MobileNetV2 backbone, the model's segmentation accuracy is 5.04% higher. Although MobileNetV2 is also lightweight, this model has a smaller computational load and a higher FPS than MobileNetV2 by 26.71 frames per second. YOLOv8 outperforms YOLOv5 in segmentation accuracy and FPS, but still lags behind SiaT-Hough in both. In summary, the SiaT-Hough model has significant average accuracy and impressive segmentation speed advantages on Mars simulation datasets, achieving a good balance between accuracy and real-time performance.

As shown in Table 2, the SiaT-Hough model achieves an average accuracy of 87.86%, outperforming SiaT-STM by 4.66% and SiaT-STM-QAM by 2.26%. Notably, its enhanced

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architecture incorporating the Hough module further improves the segmentation performance compared to the base SiaT-STM model and the SiaT-STM-QAM model with the QAM module. This indicates that the integration of the Hough module significantly boosts the segmentation accuracy. The SiaT-Hough model also achieves the highest mIoU of 78.37%, demonstrating its superior segmentation capability. The AP (PA) values are crucial for evaluating target recognition and prediction accuracy in semantic segmentation. Compared to SiaT-STM and SiaT-STM-QAM, the SiaT-Hough model shows a clear improvement in both AP (PA) and mIoU metrics. In summary, the SiaT-Hough model has significant average accuracy advantages on the Mars simulation datasets, achieving the best performance among the three models by effectively combining the STM, QAM, and Hough modules.

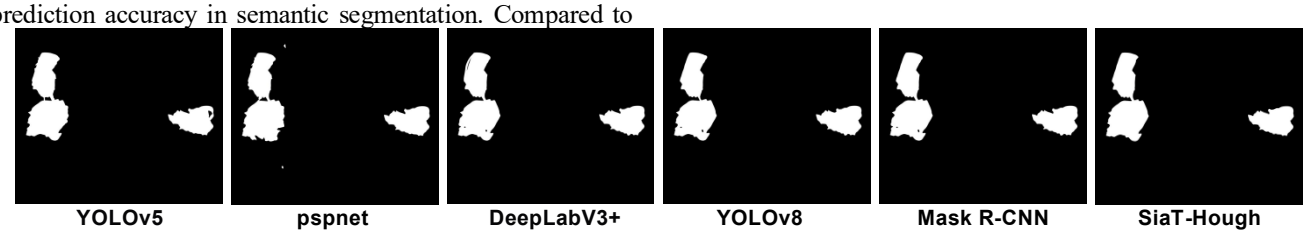

FIGURE 9.The segmentation results corresponding to six different networks.

C. SIMULATED MARS THREE-DIMENSIONAL RECONSTRUCTION

1) THREE-DIMENSIONAL RECONSTRUCTION RESULTS

First, we evaluated the collected Martian rocks to determine whether the amount of rock point cloud segmentation in the reconstructed results matched the actual amount of rock. In the reconstructed rock point cloud, in order to highlight the rock point cloud segmentation effect, we use black to represent the background, and the experimental results are shown in Figure 10. However, due to the complexity of the rock reconstruction environment in the simulation experiment and the limitation of semantic segmentation and point cloud registration accuracy, some point clouds are not correctly classified.

TABLE 3
EXPERIMENTAL RESULTS OF POSITIONING ACCURACY

Moudle	desk		texture_far	
	ATE(m)	RMSE(m)	ATE(m)	RMSE(m)
Improved ORB-SLAM2	0.0639	0.0736	0.0106	0.0116

（a） （b）

FIGURE 10.3D reconstruction results，In (b), different colors represent different stone individuals.

2) LOCALIZATION ACCURACY EVALUATION

Given Mars' complex terrain, localization accuracy is crucial for rover safety. We conducted experiments using the rgbd_dataset_freiburg1_desk(desk) and rgbd_dataset_freiburg3_structure_texture_far (texture far) public datasets to assess localization accuracy. The evaluation used absolute trajectory error (ATE) and root mean square error (RMSE). The results are shown in Table 3. ATE is calculated using equation (7).where $T_{gt,i}$ is the true 0pose at time point i, $T_{esti,i}$ is the estimated pose at time point i, $\log$ denotes the Lie algebra mapping, and N is the total number of time points. The root mean square error (RMSE) can be expressed using equation (8), where y_i is the actual value, $\hat{y}_i$ is the predicted value, and N is the sample size. In the evaluation results shown below, APE and ATE have the same meaning.

$$ATE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left\|\log(T_{gt,i}^{-1}T_{esti,i})^{\vee}\right\|_2^2} \quad (7)$$

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2} \quad (8)$$

In the simulated Mars scenario, the unmanned vehicle often encounters the situation of rock occlusion and entering the impact crater when working, so the robustness of the system needs to be analyzed in this case. For the experiment, we selected a desk dataset that simulates the descent and ascent motion into the impact crater for a period of time during the recording process. The experimental results are shown in Figure 10. During the movement in the vertical direction, the system can quickly reposition and form the predicted trajectory again. Figure 11 shows the predicted trajectory deviation of our system on the desk dataset. Figure 12 shows the ATE evaluation results.

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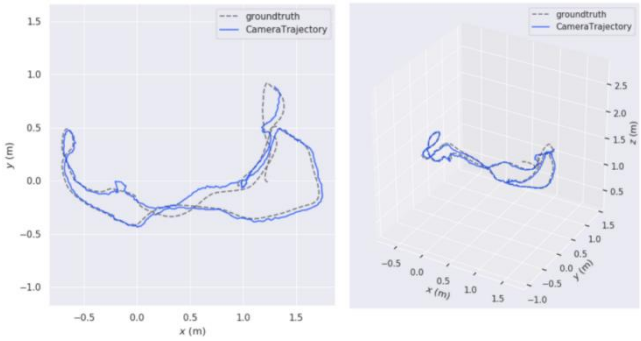
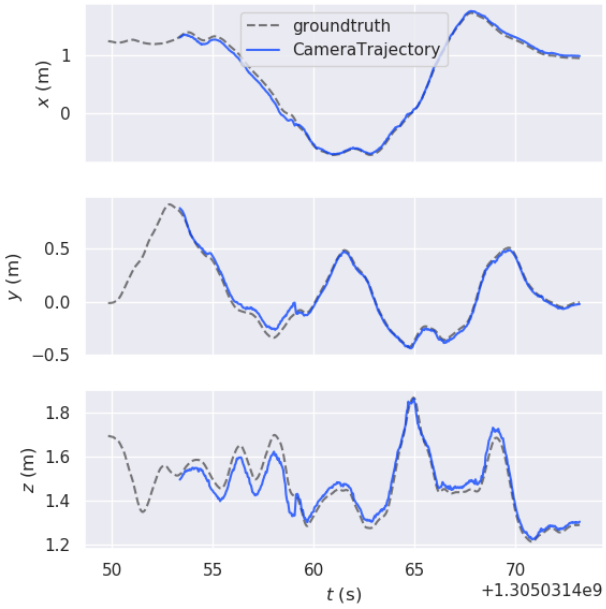
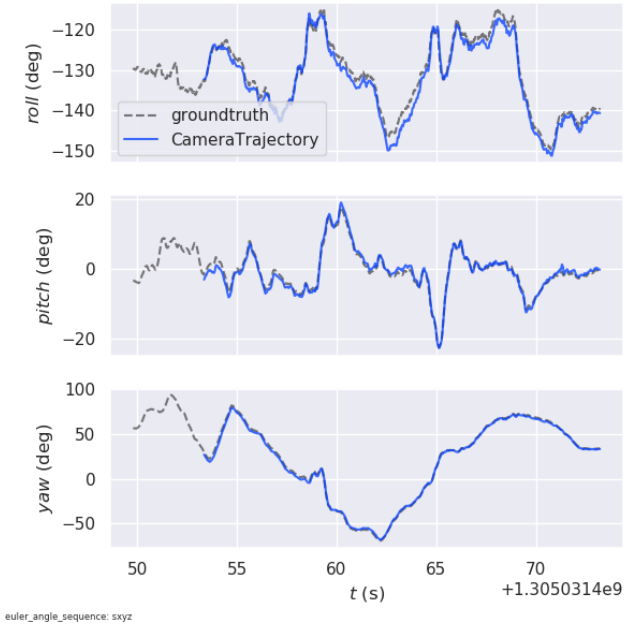


(a) (b)

FIGURE 11.desk dataset experimental results. (a) represents the trajectory diagram under the plane coordinate system; (b) represents the trajectory diagram under the space coordinate system. The dashed gray line represents the true trajectory, and the solid blue line represents the predicted trajectory of the ORB-SLAM2 system.



(a)



(b)

FIGURE 12.desk data set trajectory deviation results. (a) represents the trajectory displacement deviation diagram, and (b) represents the trajectory rotation deviation diagram.

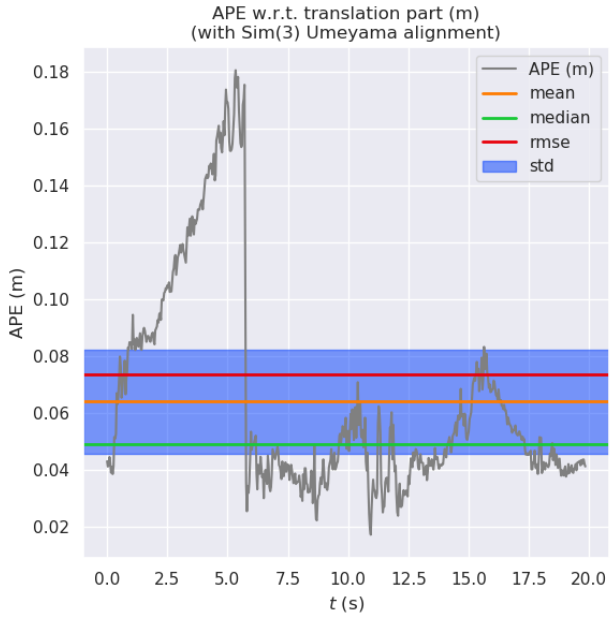


FIGURE 13.desk dataset ATE evaluation results

In the initial phase (approximately 0-6 seconds) of the ATE evaluation for the desk dataset with loops shown in Figure 13, high errors are observed, with APE values close to 0.2 meters. This is attributed to significant initialization errors and substantial camera motion errors at the start. Over time, the errors stabilize and remain below 0.08 meters, indicating a stable absolute trajectory in the subsequent period. Most absolute trajectory errors are very small, with a mean of less than 0.07 meters and a median below 0.05

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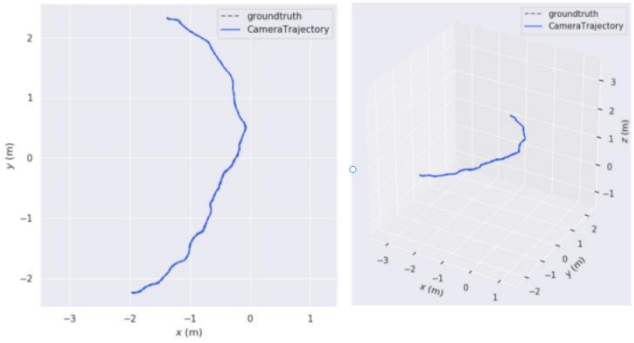
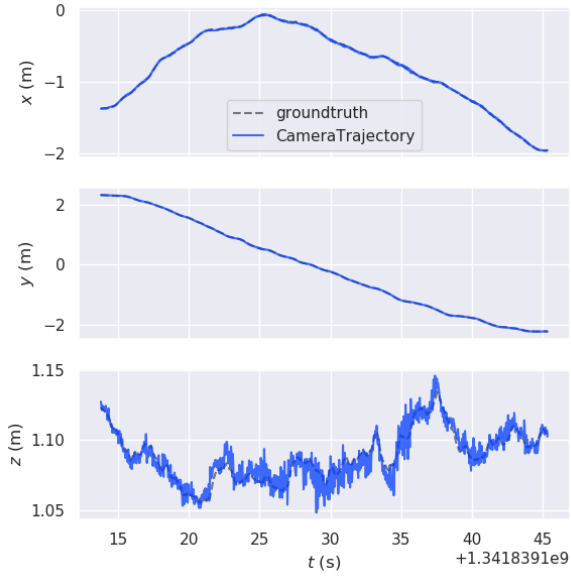
meters. The median value better represents the typical error level.

Texture_far dataset was also used for experiments, and the experimental results were shown in Figure 13. During the horizontal movement, the ORB-SLAM2 system showed the stability of positioning, and Figure 14 showed the predicted trajectory deviation of the ORB-SLAM2 system on the texture_far dataset.

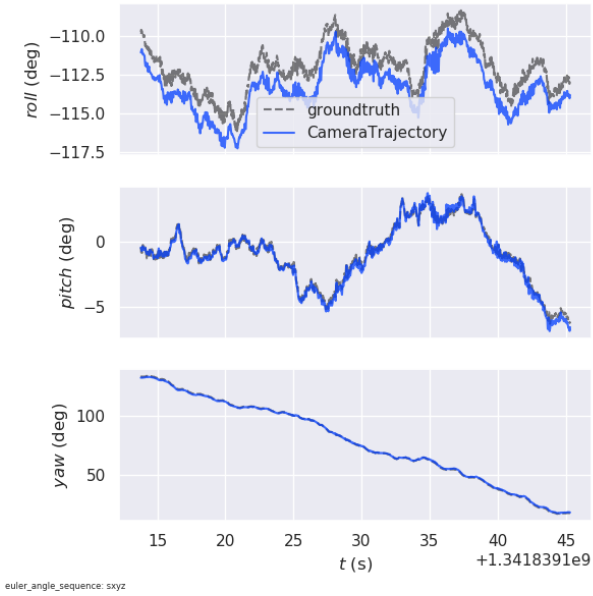


(a) (b)

FIGURE 14.texture_far Data set trajectory deviation result. (a) represents the trajectory diagram under the plane coordinate system; (b) represents the trajectory diagram under the space coordinate system



(a)



(b)

FIGURE 15.texture_far Data set trajectory deviation result. (a) represents the trajectory displacement deviation diagram, and (b) represents the trajectory rotation deviation diagram

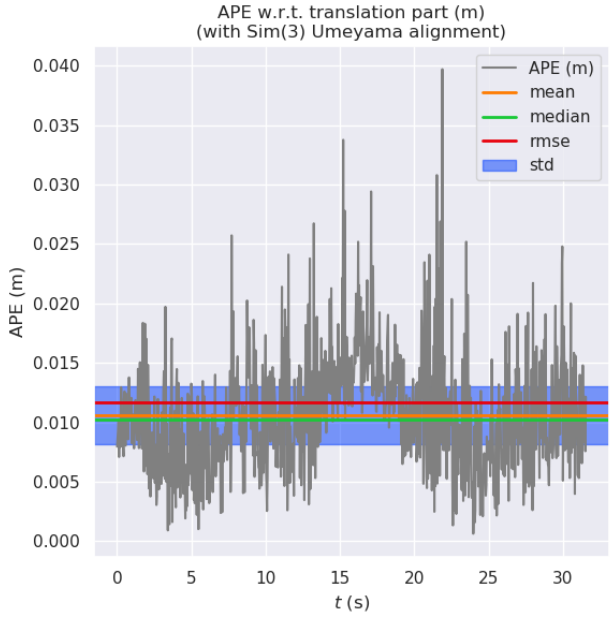


FIGURE 16.texture_far Data set ATE evaluation result

In Figure 16, the ATE evaluation results for the texture_far dataset without loops show overall stability. A few frames have errors exceeding 0.025 m, but the mean and median errors are 0.011 m. The RMSE, indicating the overall trajectory error, is 0.012 m. The standard deviation of the error is within 0.005 m, showing stable error fluctuations.

As can be seen from Table 2 and Figure 13, in this scenario, our system ATE is 0.0639m and RMSE is 0.0736m, indicating that the system has high positioning accuracy in

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TABLE 4
POINT CLOUD QUALITY RESULTS. "↑" INDICATES HIGHER VALUES ARE BETTER, WHILE "↓" INDICATES LOWER VALUES ARE BETTER.

Method	Dataset	PLCC↑	SRCC↑	RMSE↓
CLIP-PCQA (NR)	First Dataset	0.827	0.853	2.259
	Second Dataset	0.736	0.712	1.642
	Third Dataset	0.889	0.872	1.053
	Fourth Dataset	0.873	0.854	1.153
	Fifth Dataset	0.935	0.915	0.824
average accuracy		0.852	0.841	1.386

most cases. As can be seen from Figure 16, ATE and RMSE of the predicted trajectory are 0.0106m and 0.0116m, respectively. RMSE values remain around 0.010 m in the whole time range, and the median error is also relatively stable, slightly lower than the average error, indicating that most errors are concentrated in a lower range. Therefore, it is proved that our system has high robustness in simulation environment. This reflects better graphing effects.

3) POINT CLOUD QUALITY EVALUATION

To fully assess the quality of 3D - reconstructed point clouds, we conduct experiments on five Mars - scene datasets from our simulation platform. Three widely - used metrics quantify point - cloud quality: Pearson Linear Correlation Coefficient (PLCC): As per equation (9), PLCC is calculated. Here, x_i and y_i are the values of the i point in the ground-truth and reconstructed point clouds. $\bar{x}$ and $\bar{y}$ are their respective means, and n is the total number of points. Spearman Correlation Coefficient (SRCC) and Root Mean Square Error (RMSE): As shown in equation (10), SRCC and RMSE are computed. r_i and s_i represent the ranks of the i point in the ground-truth and reconstructed point clouds.

$$PLCC = \frac{\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i-\bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i-\bar{y})^2}} \quad (9)$$

$$SRCC = \frac{\sum_{i=1}^{n}(r_i-\bar{r})(s_i-\bar{s})}{\sqrt{\sum_{i=1}^{n}(r_i-\bar{r})^2}\sqrt{\sum_{i=1}^{n}(s_i-\bar{s})^2}} \quad (10)$$

The experimental results are shown in Table 4. By introducing visual language alignment and distribution prediction, we assessed the average quality of five point cloud images using the CLIP-PCQA[25] method. PLCC reached 0.852 and SRCC reached 0.841, indicating a strong linear and monotonic correlation between reconstructed point clouds and real point clouds. The RMSE also reached a smaller 1.386, indicating a smaller average error between the reconstructed point cloud and the real point cloud. Can prove that point cloud quality has a good effect. These indicators show that the reconstructed point cloud has high similarity and accuracy with the real point cloud, and can meet the requirements of 3D reconstruction in Mars exploration missions.

TABLE 5
ROCK COUNTING RESULTS

SAMPLE	PREDICTED VALUE	ACTUAL VALUE	RELATIVE ERROR Δ (%)
1	6	6	0
2	8	7	12.50
3	15	14	6.67
4	28	24	14.29
5	31	28	9.68
MEAN	\	\	8.63

4) MARS ROCK COUNTING EXPERIMENT

Accurate counting of rocks on the surface of Mars is of great significance for geological research and path planning during Mars exploration missions. This experiment aims to verify the validity and accuracy of the Mars rock counting method based on semantic segmentation and three-dimensional point cloud image reconstruction. The objects in the image are divided into rock and background. Through semantic segmentation and point cloud processing, the improved ORB-SLAM2 system is able to accurately segment and count rocks on the Martian surface. The experiment is shown in Table 5, where relative error is used to quantitatively describe the counting results. The relative error δ can be expressed using the formula (11), where Np represents the predicted value and Nt represents the true value. The results show that the average relative error of rock counting is 8.63%, which shows that the method has good performance in the Mars rock counting experiment and can provide technical support for the Mars exploration mission.

$$\delta = \frac{|N_p - N_t|}{N_t} \times 100\% \quad (11)$$

IV. CONCLUSIONS

Aiming at the key requirements of environmental simulation and perception algorithms in Mars exploration missions, this research constructs a high-fidelity digital twin platform and an efficient case segmentation model, which provides technical support for autonomous navigation and scientific exploration of Mars probes. On this simulation platform, the ORB-SLAM2 system integrating SiaT-Hough semantic segmentation module is used to reconstruct the Mars simulation environment in three dimensions and count the rocks, which provides an exploratory method and basis for the future Mars exploration.

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The multi-scale Mars environment dynamic simulation was realized based on NVIDIA Isaac Sim's programmed terrain generation method, and three-dimensional semantic reconstruction was carried out in the simulation scene combined with the improved ORB-SLAM2 system. The average absolute trajectory error of the system positioning accuracy was 0.0639m and 0.0106m on the two public data sets. The reliability of location and reconstruction is verified. The Pearson linear correlation coefficient of the mean point cloud mass of the Mars rendering dataset reached 0.852 in the point cloud quality evaluation, demonstrating the feasibility of the improved ORB-SLAM2 system for rock counting in the Mars environment. The SiaT-Hough semantic segmentation model is proposed to segment Mars rock images, and the average accuracy of semantic segmentation can reach 87.86%, which is significantly better than the mainstream comparison model. The average relative error in the rock counting experiment is only 8.63%, which verifies the feasibility of Mars rock counting. Experiments show that the lightweight network combined with geometric prior and high-precision simulation environment can effectively improve the safety of Mars exploration missions.

However, the current research still has some limitations: the light and atmospheric scattering models in the simulation environment have not fully covered the extreme conditions of Mars, which may affect the generalization ability of the simulation to the real world; The consistency of edge segmentation of SiaT-Hough model in dense rock scenes still has room for optimization, and the effect of real-time deployment on edge computing platform needs to be further verified. In the future, the multi-physics coupling simulation and adaptive feature fusion mechanism will integrate more rock geometry and texture, create Martian soil terrain texture, and improve the program distribution method of rocks, further improve the authenticity of the simulated environment and improve the robustness of the system in the dynamic Mars environment.

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